CHANDIMA BANDARA

Undergraduate | BSc (Hons) in IT specializing in Data Science

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+94777284169   My Portfolio

PROFILE

Highly motivated IT undergraduate at Sri Lanka Institute of Information Technology with skills in Python, Java, HTML, CSS, and SQL. Seeking a Software Engineering internship or entry-level role to apply coding, analytical, and problem-solving skills in real-world software development.

EDUCATION

2024 – present Malabe, Sri Lanka	**BSc (Hons) in Information Technology** *Sri Lanka Institute of Information Technology* (*SLIT*)
2022 Dodamgaslandha, Sri Lanka	**Sir John Kotelawala C.C.** I completed my Advanced Level (AL) education with a focus on the technology stream.
2019 Kurunagala, Sri Lanka	**Namal Anga M.V.** I completed my Ordinary Level (OL)

PROJECTS

AI-Powered Resume Screening System

- **Technologies**: Python, Gradio, pdfplumber, PyPDF2, BeautifulSoup, Hugging Face Serverless API, Qwen 2.5 7B Instruct, Plotly
- Developed an intelligent, **RAG-based** recruitment tool to automate the candidate evaluation process against specific job requirements.
- Implemented PDF parsing and automated web scraping to extract resume text and verify live project data from embedded portfolio links.
- Built an interactive dashboard to display structured evaluations, including visual match percentages, skill breakdowns, critical gaps, and hiring recommendations.
- Integrated the Qwen 2.5 7B Instruct model via the Hugging Face API for context-aware AI scoring and bias reduction based on verified skills.
- Links: GitHub Repository | Live Demo

Student Concern Management System (SCMS)

- **Technologies:** Java 21, Spring Boot, Spring Security, Spring Data JPA, Thymeleaf, Microsoft SQL Server, Google Gemini API
- Collaborated with a real client to gather requirements and develop a role-based concern management system for Students, Admins, and Owners.
- Implemented secure authentication with email OTP verification, password recovery, BCrypt encryption, and role-based access control.
- Built workflow-driven concern tracking with draft saving, evidence uploads, status management, admin responses, and meeting scheduling.
- Integrated Google Gemini API for AI-powered moderation of community forum posts and replies.
- Developed a centralized notification system supporting direct and broadcast communication.
- **Links:** GitHub Repository | Live Demo | Project Details

Student Concern Management System (SCMS) – Mobile App

- **Technologies:** React Native, Expo, Node.js, Express.js, MongoDB (Atlas), Mongoose, JWT, Bcryptjs, Multer, Nodemailer, Axios
- Developed a full-stack, multi-role mobile application to streamline communication between students and academic administration.

- Architected a MERN stack system using a clean MVC pattern, featuring dedicated dashboards and workflows for **Student, Admin/Consultant, and Owner roles**.
- Implemented a real-time concern tracking pipeline with status updates (Pending → In Progress → Resolved), along with automated email notifications.
- Developed a secure medical triage module with file upload support and restricted access control for sensitive health-related documentation.
- Integrated institutional oversight features such as performance analytics, PDF/Excel report generation, and broadcast announcement tools.
- **Links:** GitHub Repository | Project more details

Semantic Priority Analysis – Campus LMS Support Triage System
- **Technologies:** Python, Jupyter Notebook, Pandas, Scikit-learn, XGBoost, Transformers (DistilBERT, BART MNLI), NLTK, SMOTE, TF-IDF, Plotly, Seaborn
- Developed an end-to-end NLP and machine learning pipeline to automatically classify LMS student support reports into priority levels, reducing manual triage effort and delays.
- Designed a hybrid labeling strategy combining rule-based keyword extraction, zero-shot semantic classification (BART MNLI), and synthetic data augmentation to generate reliable labels from a 1005-record dataset.
- Applied advanced preprocessing techniques including text cleaning, TF-IDF vectorization, and SMOTE-based oversampling to handle severe class imbalance.
- Built and compared multiple machine learning models such as XGBoost, SVM, and Random Forest, and fine-tuned a DistilBERT transformer model for improved classification performance.
- Exported trained model artifacts and implemented a reusable predict_priority function capable of returning predicted classes with confidence scores for new incoming reports.
- **Links:** GitHub Repository | Live Demo

Student Mental Health Early Detection System
- **Technologies:** Python, Jupyter Notebook, Pandas, Scikit-learn, Keras/TensorFlow, Multilayer Perceptron (MLP), Exploratory Data Analysis (EDA)
- Developed an end-to-end machine learning pipeline to detect early signs of mental health risk in students based on academic performance and stress-related factors.
- Led the data preprocessing phase, including data cleaning, transformation, and outlier removal to ensure high-quality structured input from a Kaggle dataset.
- Designed a comparative modeling pipeline evaluating multiple algorithms such as KNN, SVM, Logistic Regression, Decision Trees, and Random Forest.
- Trained and optimized a Multilayer Perceptron (MLP) neural network, achieving a peak accuracy of **90.91%**.
- Generated detailed evaluation metrics including confusion matrix, accuracy trends, and loss curves to validate model performance.
- **Links:** GitHub Repository | Live demo

VeloRent - Web-Based Car Rental System
- **Technologies:** Java 17, Spring Boot, Microsoft SQL Server, HTML, CSS, JavaScript, Thymeleaf, Maven, RESTful APIs, Git
- Developed a comprehensive web application to automate the vehicle rental lifecycle, enabling real-time booking and secure user authentication.
- Acted as Lead Developer for system security, implementing user authentication, registration, Two-Factor Authentication (2FA), and password recovery mechanisms.
- Implemented Role-Based Access Control (RBAC) to manage secure permissions across **Customers, Owners, Fleet Managers, and Admin roles**.
- Designed and developed complex backend logic and dashboards for Fleet Managers to monitor real-time vehicle availability, active rentals, and return status.
- Collaborated within a team to integrate key modules, including secure payment processing, vehicle inventory management, and dynamic pricing rules.
- **Links:** GitHub Repository

SKILLS

Technical Skills
Python, Java, Spring Boot, MERN, MongoDB, MYSQL, Machine Learning, Git

Soft Skills
Analytical, Teamwork, Communication, Hard Work, Leadership